Modeling the impact of non-pharmaceutical interventions and vaccination on COVID-19-like outbreaks: a SEIRD compartmental approach

Author: [SEU NOME AQUI] — Universidade Federal Rural de Pernambuco (UFRPE)

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Abstract

Compartmental models remain the workhorse of theoretical epidemiology, providing a tractable link between mechanistic assumptions about disease transmission and observable outbreak trajectories. Here we implement a Susceptible-Exposed-Infectious-Recovered-Deceased (SEIRD) model, parameterized with literature-derived estimates for early SARS-CoV-2 circulation (R0 = 2.5, incubation period 5.2 days, infectious period 7.0 days, case fatality rate 1%), and use it to quantify the effect of two public-health interventions in a closed population of one million individuals: a temporary 60% reduction in transmission (a lockdown-style non-pharmaceutical intervention, NPI) and a constant-rate vaccination campaign. A temporary NPI applied for 90 days delays the epidemic peak by roughly 90 days but barely reduces its height (131,904 vs. 130,940 peak infectious individuals) once lifted, because the susceptible pool is not depleted while the intervention is active — a numerical illustration of why "flattening the curve" without sustained suppression mainly buys time rather than reducing final severity. In contrast, a vaccination campaign starting on day 60 at 1%/day of the remaining susceptible pool reduces the attack rate from 89.3% to 40.2%, the death toll by 55%, and peak prevalence by 63%. A sensitivity analysis over R0 in [1.1, 4.9] confirms the expected threshold behavior at R0 = 1 and reproduces the classical final-size relation. Full source code is openly available (Code Availability).

Keywords: SEIRD model, compartmental epidemiology, COVID-19, non-pharmaceutical interventions, vaccination, basic reproduction number

1. Introduction

Since the foundational work of Kermack and McKendrick (1927), compartmental models — partitioning a population into mutually exclusive epidemiological classes and describing transitions between them with ordinary differential equations (ODEs) — have provided the principal quantitative language for understanding how an infectious disease spreads through a population, and for evaluating, in silico, the likely impact of interventions before they are deployed at scale. The COVID-19 pandemic renewed widespread interest in this class of models, as governments worldwide relied on SEIR-type projections to justify lockdowns, school closures, and vaccination prioritization strategies.

The simplest model in this family, the SIR model, assumes that infected individuals become infectious immediately upon exposure. This is a poor approximation for diseases such as COVID-19, which have a non-trivial incubation (latent) period during which an individual is infected but not yet infectious. The SEIR extension introduces an Exposed (E) compartment to capture this delay, and is the standard model used for SARS-CoV-2. Because a policy-relevant question — how many deaths does an intervention avert? — cannot be answered without an explicit fatality process, we further extend the model to SEIRD, adding a Deceased (D) compartment fed by a case-fatality branch off the Infectious compartment, and a V (vaccinated) bookkeeping compartment to track vaccine-derived immunity separately from infection-derived immunity.

2. Methods

2.1 Model structure

We model a closed population of size N partitioned into six mutually exclusive compartments: Susceptible (S), Exposed (E), Infectious (I), Recovered (R), Deceased (D), and Vaccinated (V), with N = S + E + I + R + D + V constant over time. The dynamics are governed by the system of ODEs:

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dS/dt = -β(t)·S·I/N − v(t)·S
dE/dt =  β(t)·S·I/N − σ·E
dI/dt =  σ·E − γ·I
dR/dt =  (1−μ)·γ·I
dD/dt =  μ·γ·I
dV/dt =  v(t)·S
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where $\beta(t)$ is the (possibly time-varying) transmission rate; σ = 1/(incubation period) is the rate at which exposed individuals become infectious; γ = 1/(infectious period) is the recovery/removal rate; μ is the case fatality rate; and v(t) is the per-capita vaccination rate applied to the susceptible pool. The basic reproduction number is $R0 = \beta/\gamma$ in the absence of interventions. NPIs are modeled as a fractional reduction in β active over a fixed window; vaccination as a constant daily removal rate applied to S, moving individuals directly into V without passing through infection.

Separating V from R is a deliberate modeling choice: without it, a naive attack rate computed as 1 − S(final)/N would count vaccinated, never-infected individuals as though they had been infected, conflating vaccine coverage with disease burden. We avoid this structurally rather than by post-hoc correction.

2.2 Parameterization

Parameter values were taken from early-pandemic epidemiological studies of SARS-CoV-2 (Table 1), representing a Wuhan-like ancestral strain without variant-specific transmissibility increases.

Parameter	Value	Source
R0	2.5	WHO situation reports (2.0-3.5)
Incubation period ($1/\sigma$)	5.2 days	Lauer et al., Ann Intern Med 2020
Infectious period ($1/\gamma$)	7.0 days	Nishiura et al., Int J Infect Dis 2020
Case fatality rate (μ)	1.0%	WHO crude early estimate
Population (N)	1,000,000	Closed population
Initial conditions	E0=20, I0=5	Arbitrary seeding

Table 1. Baseline epidemiological parameters.

2.3 Scenarios

Three scenarios were simulated over 365 days: (1) Baseline — no intervention, constant β; (2) NPI — a 60% reduction in β active from day 30 to day 120; (3) Vaccination — a campaign starting day 60, vaccinating 1% of the remaining susceptible pool per day, with no NPI. A fourth analysis swept R0 from 1.1 to 4.9 (step 0.2) with no interventions.

2.4 Numerical implementation

The ODE system was integrated in Python 3 using scipy.integrate.solve_ivp with the explicit Runge-Kutta 4(5) method (RK45), maximum step size 1 day, relative tolerance 1e-8, absolute tolerance 1e-6. Source code is organized as a reusable SEIRDModel class accepting pluggable Intervention and Vaccination objects.

3. Results

3.1 Baseline outbreak dynamics

Without intervention, the outbreak peaks at day 115.6, with 131,904 individuals (13.2% of the population) simultaneously infectious, and an eventual attack rate of 89.3% and 8,926 deaths (Fig. 1, left; Fig. 2). This is consistent with the final-size relation for SIR/SEIR models at R0 = 2.5.

3.2 Effect of a temporary non-pharmaceutical intervention

A 60% transmission reduction from day 30 to day 120 delays the epidemic peak from day 115.6 to day 205.5 (Fig. 1, middle; Fig. 2, blue). However, because the intervention is lifted before the susceptible pool is meaningfully depleted, transmission rebounds once β returns to baseline: peak prevalence is reduced by only 0.7%, and the final attack rate (89.2%) and death toll (8,920) are essentially unchanged. This reproduces a well-documented result: a temporary NPI that is not sustained delays, but does not by itself prevent, a large outbreak.

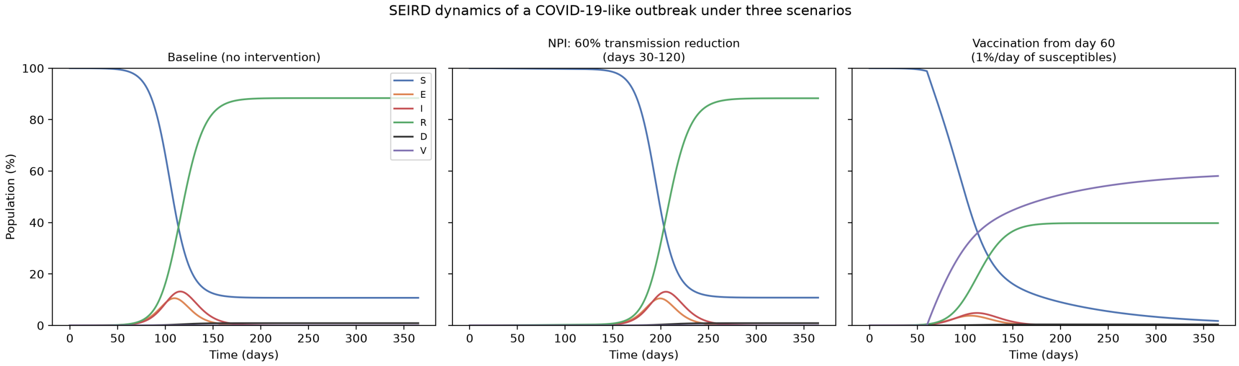

Figure 1. SEIRD compartment trajectories (% of population) over 365 days under three scenarios: no intervention (left), a 60% transmission reduction active from day 30-120 (middle), and a vaccination campaign starting day 60 at 1%/day of the susceptible pool (right).

3.3 Effect of a vaccination campaign

A vaccination campaign starting day 60, immunizing 1% of the remaining susceptible pool per day, substantially changes outbreak severity (Fig. 1, right; Fig. 2, green): peak infectious prevalence falls from 131,904 to 48,421 (a 63.3% reduction), the attack rate among those never vaccinated falls from 89.3% to 40.2%, and deaths fall from 8,926 to 4,017 (a 55.0% reduction), with 58.1% of the population vaccinated by day 365. Unlike the temporary NPI, vaccination permanently removes individuals from the susceptible pool.

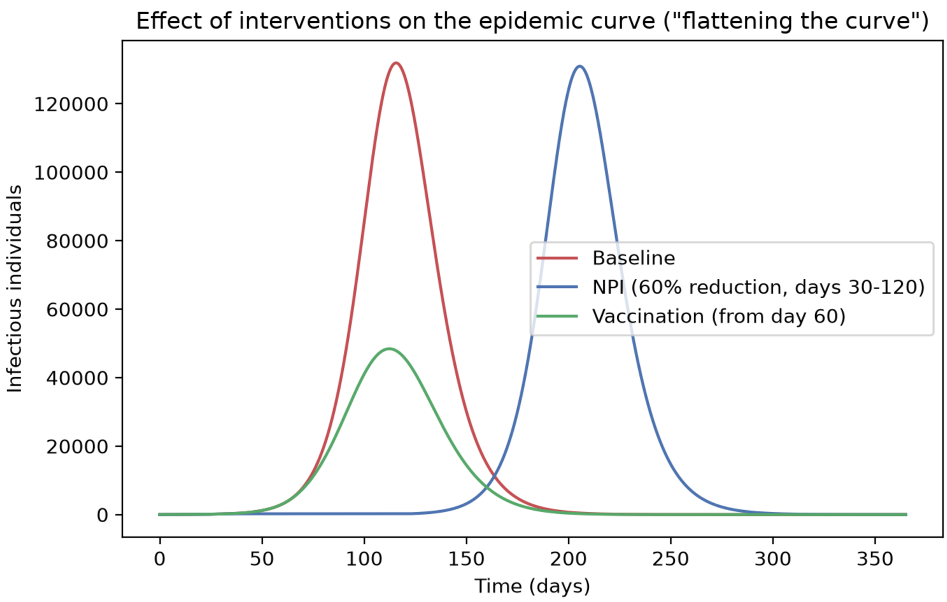

Figure 2. Infectious compartment (absolute count) over time across the three scenarios, illustrating that the temporary NPI shifts the peak in time without materially reducing its height, whereas vaccination reduces peak height directly.

3.4 Sensitivity to R0

Sweeping R0 from 1.1 to 4.9 (Fig. 3) shows the expected threshold behavior: below R0 = 1 no self-sustaining epidemic occurs, while peak prevalence and final attack rate both rise steeply for R0 just above 1 and then saturate (attack rate rises from 1.2% at R0 = 1.1 to 39.9% at R0 = 1.3, but only from 96.6% at R0 = 3.5 to 99.2% at R0 = 4.9), consistent with the concave final-size relation.

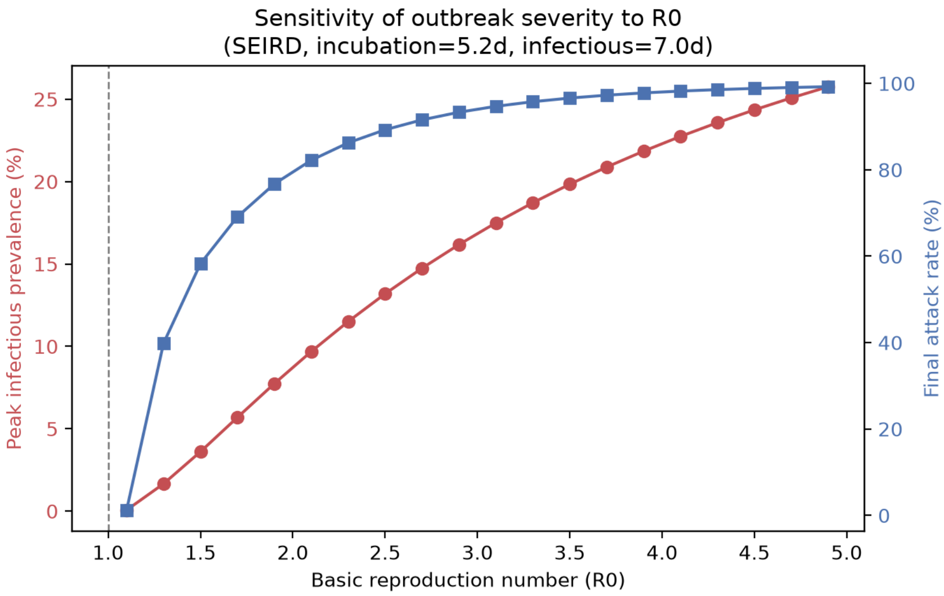

Figure 3. Sensitivity of peak infectious prevalence (red, left axis) and final attack rate (blue, right axis) to R0. The dashed line marks the epidemic threshold R0 = 1.

4. Discussion

The results illustrate a central lesson of compartmental epidemiology: the value of a non-pharmaceutical intervention is not measured by its peak-reduction effect while active, but by what it buys time for — expanded healthcare capacity, treatment development, or, as modeled here, a vaccination campaign. A temporary NPI on its own, absent a mechanism that permanently removes individuals from the susceptible pool, delays but does not prevent a large outbreak.

Several simplifying assumptions limit realism: homogeneous mixing (ignoring age structure, spatial and contact-network effects); permanent immunity over the one-year horizon (ignoring waning and reinfection); a constant, population-uniform case fatality rate; and instantaneously effective, untargeted vaccination. Extending the model to a multi-group SEIRD system, or coupling it to a spatial/network (e.g. cellular-automaton) structure, would relax the homogeneous-mixing assumption and is a natural direction for future work. Despite these simplifications, the model reproduces well-established patterns (the final-size relation, the R0 = 1 threshold, and the delay-vs-suppression distinction) and provides a compact, open, reusable tool for exploring what-if scenarios.

5. Conclusion

We implemented and validated a SEIRD model of a COVID-19-like outbreak and compared a temporary NPI against a sustained vaccination campaign. Under realistic parameters, a temporary 60% transmission reduction delays the peak by ~3 months but leaves the final attack rate and death toll essentially unchanged, whereas vaccination reduces peak prevalence by 63%, the attack rate by roughly half, and deaths by 55%. A sensitivity analysis confirms the model reproduces the epidemic threshold at R0 = 1 and the classical final-size relation. All code is openly available for reuse.

Code and Data Availability

Source code (Python 3, SEIRDModel class), generated CSV time series, and all figures are available at:
GitHub: [ADICIONAR LINK DO SEU REPOSITÓRIO GITHUB AQUI]

AI Usage Disclosure

Portions of this project's code and this article were developed with assistance from Claude Code (Anthropic, model claude-sonnet-5), an AI coding agent, under direct guidance and review by the author. The author specified the disease (COVID-19), the scenarios to compare (baseline, NPI, vaccination), and reviewed and validated the model equations, parameter sources, and all numerical outputs. A representative prompt was: "implement a SEIRD model for COVID-19 with a baseline, an NPI scenario, and a vaccination scenario, using literature-based parameters, and generate comparison figures and a sensitivity analysis over R0."
Prompt/session link: [ADICIONAR LINK/EXPORT DA CONVERSA COM A IA AQUI, SE DISPONÍVEL]

References

1. Kermack, W. O. & McKendrick, A. G. A contribution to the mathematical theory of epidemics. Proc. R. Soc. Lond. A 115, 700-721 (1927).
2. Keeling, M. J. & Rohani, P. Modeling Infectious Diseases in Humans and Animals. Princeton University Press (2008).
3. World Health Organization. COVID-19 Situation Reports (2020).
4. Lopez, L. & Rodo, X. A modified SEIR model to predict the COVID-19 outbreak in Spain and Italy. Results Phys. 21, 103746 (2021).
5. Giordano, G. et al. Modelling the COVID-19 epidemic and implementation of population-wide interventions in Italy. Nat. Med. 26, 855-860 (2020).
6. Lauer, S. A. et al. The incubation period of COVID-19 from publicly reported confirmed cases. Ann. Intern. Med. 172, 577-582 (2020).
7. Nishiura, H., Linton, N. M. & Akhmetzhanov, A. R. Serial interval of novel coronavirus (COVID-19) infections. Int. J. Infect. Dis. 93, 284-286 (2020).
8. Flaxman, S. et al. Estimating the effects of non-pharmaceutical interventions on COVID-19 in Europe. Nature 584, 257-261 (2020).
9. Chang, S. et al. Mobility network models of COVID-19 explain inequities and inform reopening. Nature 589, 82-87 (2021).